

STAT-437: APPLIED STATISTICAL METHODS

Assignment: 1

Due: September 18, 2026 at 10:00PM

Total=40 Marks

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Instructions: (i) For Question 2 Please ensure that all answers and relevant R outputs are presented in the main section of your submission. The corresponding R code should be included in the Appendix and submitted as a source file either a script file (.r) or a markdown file (.rmd). Kindly note that late submissions will not be accepted.

Question 1: A real-estate company wants to investigate factors that affect the selling price of a house. The company considers two explanatory variables:

$\mathbf{x}_1$ = house size (measured in hundreds of square feet);

$\mathbf{x}_2$ = number of bedrooms;

$\mathbf{y}$ = selling price (measured in thousands of dollars).

The company uses the multiple linear regression model

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \epsilon_i, \quad i = 1, 2, \dots, n$$

where $E(\epsilon_i) = 0$ and $Var(\epsilon_i) = \sigma^2$.

In matrix form,

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

where $\boldsymbol{\beta}^T = (\beta_0, \beta_1, \beta_2)$.

(i) Write the design matrix $\mathbf{X}$ for this multiple linear regression model.

(ii) The least-squares estimator of $\boldsymbol{\beta}$ is obtained by minimizing

$$S(\boldsymbol{\beta}) = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$$

Show that

$$S(\boldsymbol{\beta}) = \mathbf{y}^T \mathbf{y} - 2\boldsymbol{\beta}^T \mathbf{X}^T \mathbf{y} + \boldsymbol{\beta}^T \mathbf{X}^T \mathbf{X} \boldsymbol{\beta}$$

(iii) Differentiate $S(\boldsymbol{\beta})$ with respect to $\boldsymbol{\beta}$ and show that

$$\frac{\partial S}{\partial \boldsymbol{\beta}} = -2\mathbf{X}^T \mathbf{y} + 2\mathbf{X}^T \mathbf{X} \boldsymbol{\beta}$$

(iv) By setting the derivative of $\frac{\partial S}{\partial \boldsymbol{\beta}}$ equal to zero, show that the least-squares estimator is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

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(v) Show that $\hat{\beta}$ is an unbiased estimator of β .

(vi) Suppose the fitted regression equation is

$$\hat{y} = 40 + 2.5x_1 + 8x_2$$

Interpret the coefficients $\hat{\beta}_1 = 2.5$ and $\hat{\beta}_2 = 8$ in the context of the problem.

(vii) Use the fitted model $\hat{y} = 40 + 2.5x_1 + 8x_2$ to predict the selling price of a house that has 2,000 square feet and 3 bedrooms.

Question 2: Teengamb Dataset

The dataset **teengamb**, available in the R package **faraway**, relates to a study of teenage gambling in Britain. Fit a multiple linear regression model with gambling expenditure as the response variable, and sex, status, and verbal score as the predictors.

(i) What is the residual mean square (MS_{Res}) for the fitted model?

(ii) Fit a multiple linear regression model to predict gambling expenditure among teenagers using sex, status, and verbal score as predictors.

(iii) Examine the presence of multicollinearity in the fitted model using the Variance Inflation Factor (VIF).

(iv) What are the 95% confidence intervals for the coefficients of sex (β_1) and verbal score (β_3)?

(v) Does the Ordinary Least Squares (OLS) model provide a better fit to the data compared to the robust regression model?

(vi) Identify the outliers using an Ordinary Least Squares (OLS) regression model with gambling expenditure as the response and sex and status as the predictors.